

1. Which are the top three variables in your model which contribute most towards the probability of a lead getting converted?

The top three variables contributing most towards the probability of a lead getting converted are determined by the **largest absolute magnitude of their coefficients** (beta)

1. **What_is_your_current_occupation_Working Professional (beta = +2.56):** A lead who is a Working Professional has the highest positive impact on the conversion probability, significantly increasing the Log Odds.
2. **Last_Activity_Had a Talked to Lead (beta = +1.71):** A lead who has had a direct conversation with a sales rep (indicating high engagement) has the second-highest positive impact.
3. **Do_Not_Email (beta = -1.57):** While negative, its magnitude is high. If a lead selects "Do Not Email" (value 1), the probability of conversion is severely and negatively impacted, making this a highly influential variable.

2. What are the top 3 categorical/dummy variables in the model which should be focused the most on to increase the probability of lead conversion?

We focus on the dummy variables with the **largest positive coefficients**, as these represent the characteristics the sales team should target to increase conversion.

1. **What_is_your_current_occupation_Working Professional (beta = +2.56):** Focus efforts heavily on leads identified as Working Professionals.
2. **Last_Activity_Had a Talked to Lead (beta = +1.71):** Prioritize leads who have had a sales talk, as they are close to conversion.
3. **Lead_Source_Olark Chat (beta = +1.27):** Leads originating from Olark Chat (a direct, high-intent communication method) are significantly more likely to convert.

3. X Education has a period of 2 months every year during which they hire some interns. The sales team has around 10 interns allotted to them. So, during this phase, they wish to make the lead conversion more aggressive. So, they want almost all the potential leads (i.e., the customers who have been predicted as 1 by the model) to be converted and hence, want to make phone calls to as much of such people as possible. Suggest a good strategy they should employ at this stage.

Goal: Maximize conversions by ensuring almost all potential leads are contacted (High True Positive Rate). Requirement: Make phone calls to as much of such people as possible.

1. **Optimal Cut-off-** Lower the probability cut-off (e.g., from 0.35 to 0.20) which results in This increases the number of leads classified as 'Hot Leads' (Predicted 1), ensuring that few actual converters are missed (High Sensitivity).
2. **Trade-off-** Lowering the cut-off will increase the False Positive rate (making more useless calls to cold leads), but this is acceptable during an aggressive phase with intern capacity.

Conclusion: The company should use a low probability cut-off to maximize the pool of "Hot Leads" for the sales team and interns to call, prioritizing the capture of every possible conversion.

4. Similarly, at times, the company reaches its target for a quarter before the deadline. During this time, the company wants the sales team to focus on some new work as well. So, during this time, the company's aim is to not make phone calls unless it's extremely necessary, i.e., they want to minimize the rate of useless phone calls. Suggest a strategy they should employ at this stage.

Goal: Minimize the rate of useless phone calls (Minimize False Positives). Requirement: Focus on new work; only call when extremely necessary.

Conclusion: The company should use a **high probability cut-off** to narrow the focus to only the most *probable* converters, ensuring maximum efficiency and allowing the sales team to dedicate time to other tasks.